ModelIt!

Student Activity Pack

Nature's Recycling System

Why Earth Doesn't Fill Up with Dead Stuff

Grade 5 | NGSS 5-LS2-1 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why doesn't Earth fill up with dead stuff?

2 What do you think decomposers are? What do they do?

3 Draw an arrow showing how dead leaves become nutrients for plants.

4 Name one type of decomposer you know about.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Dead Organic Matter | Soil Moisture | Decomposer Activity | Nutrient Release

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Healthy Ecosystem

Normal organic matter, normal moisture

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Drought Simulation

Lock Soil Moisture to 0%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Recovery Phase

Restore moisture levels

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Temperature — Decomposers work faster in warmth

Earthworms — They shred material before bacteria process it

pH Level — Acidic soil slows certain bacteria

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why doesn't Earth fill up with dead stuff? Use evidence from your simulation to support your answers.

1 What happens to a forest ecosystem when decomposers slow down?

2 If nutrients stop being released, what happens to living plants?

3 Why do tropical forests have almost no dead material on the ground?

4 How is your backyard compost bin similar to this model?

5 What role do decomposers play in the larger matter cycle?

8 STEM Challenge: Design a Better Compost System

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YOUR ENGINEERING MISSION

Design a backyard composting system that maximizes decomposer activity year-round.

THE SCENARIO

Your school wants to compost cafeteria food waste. You need to design a system that works even in dry summer months.

Engineering Design Thinking

1. WHAT SIZE SHOULD YOUR COMPOST BIN BE?
2. HOW WILL YOU ADD MOISTURE WITHOUT OVERWATERING?
3. WHAT'S THE RIGHT MIX OF BROWN (CARBON) AND GREEN (NITROGEN) MATERIALS?
4. HOW WILL YOU KNOW WHEN COMPOST IS READY TO USE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Soil Scientists and Bioremediation Specialists use decomposers to clean up contaminated sites and restore damaged soil. They earn $75,000-$120,000/year.